

Abbreviations

Some units, terms, and symbols are not included in this list as they are defined in the text. Common abbreviations have been omitted. The titles of journals are abbreviated according to the general style of the *Index Medicus*.

≈	approximately.	IV	intravenous.
Ad	Addendum.	J	joule(s).
ADI	acceptable daily intake.	JP	Japanese Pharmacopeia.
approx	approximately.	JPE	Japanese Pharmaceutical Excipients
atm	atmosphere.	kcal	kilocalorie(s).
BAN	British Approved Name.	kg	kilogram(s).
bp	boiling point.	kJ	kilojoule(s).
BP	British Pharmacopoeia.	kPa	kilopascal(s).
BS	British Standard (specification).	L	liter(s).
BSI	British Standards Institution.	LAL	<i>Limulus</i> amoebocyte lysate.
cal	calorie(s).	LC ₅₀	a concentration in air lethal to 50% of the specified animals on inhalation.
CAS	Chemical Abstract Service.	LD ₅₀	a dose lethal to 50% of the specified animals or microorganisms.
CFC	chlorofluorocarbon.	Ld _{Lo}	lowest lethal dose for the specified animals or microorganisms.
cfu	colony-forming unit	m	meter(s).
cm	centimeter(s).	m ²	square meter(s).
cm ²	square centimeter(s).	m ³	cubic meter(s).
cm ³	cubic centimeter(s).	M	molar.
cmc	critical micelle concentration.	max	maximum.
CNS	central nervous system.	MCA	Medicines Control Agency (UK).
cP	centipoise(s).	mg	milligram(s).
cSt	centistoke(s).	MIC	minimum inhibitory concentration.
CTFA	Cosmetic, Toiletry, and Fragrance Association.	min	minute(s) <i>or</i> minimum.
d	particle diameter (d ₁₀ at 10 percentile; d ₅₀ at 50 percentile; d ₉₀ at 90 percentile).	mL	milliliter(s).
D&C	designation applied in USA to dyes permitted for use in drugs and cosmetics.	mm	millimeter(s).
DoH	Department of Health (UK).	mM	millimolar.
DSC	differential scanning calorimetry.	mm ²	square millimeter(s).
EC	European Community.	mm ³	cubic millimeter(s).
e.g.	<i>exempli gratia</i> , 'for example'.	mmHg	millimeter(s) of mercury.
EINECS	European Inventory of Existing Commercial Chemical Substances.	mmol	millimole(s).
<i>et al</i>	<i>et alii</i> , 'and others'.	mN	millinewton(s).
EU	European Union.	mol	mole(s).
FAO	Food and Agriculture Organization of the United Nations.	mp	melting point.
FAO/WHO	Food and Agriculture Organization of the United Nations <i>and</i> the World Health Organization.	mPa	millipascal(s).
FCC	Food Chemicals Codex.	MPa	megapascal(s).
FDA	Food and Drug Administration of the USA.	μg	microgram(s).
FD&C	designation applied in USA to dyes permitted for use in foods, drugs, and cosmetics.	μm	micrometer(s).
FFBE	Flat face beveled edge.	N	newton(s) <i>or</i> normal (concentration).
g	gram(s).	NIR	near-infrared.
GMP	Good Manufacturing Practice.	nm	nanometer(s).
GRAS	generally recognized as safe by the Food and Drug Administration of the USA.	o/w	oil-in-water.
HC	hydrocarbon.	o/w/o	oil-in-water-in-oil.
HCFC	hydrochlorofluorocarbon.	Pa	pascal(s).
HFC	hydrofluorocarbon.	pH	the negative logarithm of the hydrogen ion concentration.
HIV	human immunodeficiency virus.	PhEur	European Pharmacopeia.
HLB	hydrophilic-lipophilic balance.	pK _a	the negative logarithm of the dissociation constant.
HSE	Health and Safety Executive (UK).	pph	parts per hundred.
i.e.	<i>id est</i> , 'that is'.	ppm	parts per million.
IM	intramuscular.	psia	pounds per square inch absolute.
INN	International Nonproprietary Name.	R	reflectance.
IP	intraperitoneal.	RDA	recommended dietary allowance (USA).
ISO	International Organization for Standardization.	rpm	revolutions per minute.
IU	International Units.	s	second(s).
		SC	subcutaneous.
		SEM	scanning electron microscopy <i>or</i> scanning electron microphotography.

SI	Statutory Instrument <i>or</i> Système International d'Unités (International System of Units).	USP–NF	The United States Pharmacopeia National Formulary.
TD _{Lo}	lowest toxic dose for the specified animals or microorganisms.	UV	ultraviolet.
TPN	total parental nutrition.	v/v	volume in volume.
TWA	time weighted average.	v/w	volume in weight.
UK	United Kingdom.	WHO	World Health Organization.
US <i>or</i> USA	United States of America.	w/o	water-in-oil.
USAN	United States Adopted Name.	w/o/w	water-in-oil-in-water.
USP	The United States Pharmacopeia.	w/v	weight in volume.
		w/w	weight in weight.

Units of Measurement

The information below shows imperial to SI unit conversions for the units of measurement most commonly used in the *Handbook*. SI units are used throughout with, where appropriate, imperial units reported in parentheses.

Area

1 square inch (in²) = 6.4516×10^{-4} square meter (m²)
1 square foot (ft²) = 9.29030×10^{-2} square meter (m²)
1 square yard (yd²) = 8.36127×10^{-1} square meter (m²)

Density

1 pound per cubic foot (lb/ft³) = 16.0185 kilograms per cubic meter (kg/m³)

Energy

1 kilocalorie (kcal) = 4.1840×10^3 joules (J)

Force

1 dyne (dynes) = 1×10^{-5} newton (N)

Length

1 angstrom (Å) = 10^{-10} meter (m)
1 inch (in) = 2.54×10^{-2} meter (m)
1 foot (ft) = 3.048×10^{-1} meter (m)
1 yard (yd) = 9.144×10^{-1} meter (m)

Pressure

1 atmosphere (atm) = 0.101325 megapascal (MPa)

1 millimeter of mercury (mmHg) = 133.322 pascals (Pa)

1 pound per square inch (psi) = 6894.76 pascals (Pa)

Surface tension

1 dyne per centimeter (dyne/cm) = 1 millinewton per meter (mN/m)

Temperature

Celsius (°C) = $(1.8 \times ^\circ\text{C}) + 32$ Fahrenheit (°F)

Fahrenheit (°F) = $(0.556 \times ^\circ\text{F}) - 17.8$ Celsius (°C)

Viscosity (dynamic)

1 centipoise (cP) = 1 millipascal second (mPa s)

1 poise (P) = 0.1 pascal second (Pa s)

Viscosity (kinematic)

1 centistoke (cSt) = 1 square millimeter per second (mm²/s)

Volume

1 cubic inch (in³) = 1.63871×10^{-5} cubic meter (m³)

1 cubic foot (ft³) = 2.83168×10^{-2} cubic meter (m³)

1 cubic yard (yd³) = 7.64555×10^{-1} cubic meter (m³)

1 pint (UK) = 5.68261×10^{-4} cubic meter (m³)

1 pint (US) = 4.73176×10^{-4} cubic meter (m³)

1 gallon (UK) = 4.54609×10^{-3} cubic meter (m³)

1 gallon (US) = 3.78541×10^{-3} cubic meter (m³)